Spring Boot

Assignment

Problem Statement:

You have been assigned to develop a **Leave Management API** for an organization's HR system using **Spring Boot**.

The application should allow employees to submit and manage leave requests through REST APIs while following Spring Boot best practices and a clean layered architecture.

The objective of this assignment is to demonstrate your understanding of Spring Boot fundamentals, including project setup, dependency injection, REST API development, validation, exception handling, configuration management, and testing.

You may use an **in-memory repository** (for example, ConcurrentHashMap) for storing data. A database is **not required** for this assignment.

Implement the application using the following package structure:

- Controller
- Service
- Repository
- Model
- DTO
- Exception
- Configuration

Implement the following REST APIs:

Method	Endpoint	Description
GET	/leaves	Retrieve all leave requests
GET	/leaves/{id}	Retrieve a leave request by ID
POST	/leaves	Create a new leave request
PUT	/leaves/{id}	Update an existing leave request

DELETE	/leaves/{id}	Delete a leave request

Each Leave Request should contain the following fields:

- id
- employeeId
- leaveType
- startDate
- endDate
- reason
- status

Functional Requirements

Your implementation should include:

- Spring Boot project setup using Maven or Gradle
- Layered architecture (Controller → Service → Repository)
- Constructor-based Dependency Injection
- DTOs for request and response models
- Bean Validation for incoming requests
- Global Exception Handling using @RestControllerAdvice
- Configuration using application.yml or application.properties
- At least one Spring Profile (for example, dev)
- Unit tests for the service layer
- REST controller tests using MockMvc

Deliverables:

Submit the following:

- Source code uploaded to a GitHub repository.
- GitHub Pull Request (PR) link.
- Branch name used for implementation.
- A `README.md` containing:
  - Project overview
  - Build and run instructions
  - API endpoint summary
  - Testing instructions
- Screenshots showing:
  - Successful application startup
  - Successful execution of the REST APIs
  - Successful execution of the test suite
- Share daily status updates through email during the assignment.

[End]